

Clustering in Data Mining is an **unsupervised learning technique** used to group similar data points into clusters such that:

- Data points **within the same cluster** are **highly similar**
- Data points **in different clusters** are **dissimilar**

Unlike classification, clustering does **not require labeled data**. It helps in discovering hidden patterns or structures in datasets.

Key Idea

Clustering tries to answer:

 *“Which data points naturally belong together?”*

Example:

Customers with similar buying behavior can be grouped into the same cluster.

Main Types of Clustering Algorithms

1. Partitioning Methods

These divide data into a fixed number of clusters.

- **K-Means**
 - Most popular algorithm
 - Assigns points to nearest centroid
- **K-Medoids (PAM)**
 - Uses actual data points as centers (more robust to noise)

2. Hierarchical Methods

Build a tree of clusters (called a dendrogram).

- **Agglomerative Clustering (Bottom-Up)**
 - Start with individual points → merge clusters
- **Divisive Clustering (Top-Down)**
 - Start with one cluster → split recursively

3. Density-Based Methods

Clusters are formed based on dense regions of data.

- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)**
 - Can detect arbitrary-shaped clusters
 - Handles noise well
- **OPTICS**
 - Similar to DBSCAN but better for varying densities

4. Model-Based Methods

Assume data is generated from a statistical model.

- **Gaussian Mixture Model (GMM)**
 - Uses probability distributions
 - Soft clustering (a point can belong to multiple clusters)

5. Grid-Based Methods

Divide data space into a grid structure.

- **STING (Statistical Information Grid)**
- **CLIQUE**
 - Useful for high-dimensional data

6. Fuzzy Clustering

Data points can belong to multiple clusters with degrees of membership.

- **Fuzzy C-Means (FCM)**



Summary Table

Category	Algorithms
Partitioning	K-Means, K-Medoids
Hierarchical	Agglomerative, Divisive
Density-Based	DBSCAN, OPTICS
Model-Based	GMM
Grid-Based	STING, CLIQUE
Fuzzy	Fuzzy C-Means



Applications

- Customer segmentation
- Image segmentation
- Anomaly detection
- Document clustering

K-Means Clustering[®]

$A1(2, 10)$, $A2(2, 5)$,
 $A3(8, 4)$, $B1(5, 8)$,
 $B2(7, 5)$, $B3(6, 4)$,
 $C1(1, 2)$, $C2(4, 9)$

Solved
Example



K-Means Clustering – Solved Example

- Suppose that the data mining task is to cluster points into three clusters,
- where the points are
- $A1(2, 10)$, $A2(2, 5)$, $A3(8, 4)$, $B1(5, 8)$, $B2(7, 5)$, $B3(6, 4)$, $C1(1, 2)$, $C2(4, 9)$.
- The distance function is Euclidean distance.
- Suppose initially we assign $A1$, $B1$, and $C1$ as the center of each cluster, respectively.

K-Means Clustering – Solved Example

Initial Centroids:

A1: (2, 10)

B1: (5, 8)

C1: (1, 2)

Data Points			Distance to						Cluster	New Cluster
			2	10	5	8	1	2		
A1	2	10	0							
A2	2	5	5							
A3	8	4								
B1	5	8								
B2	7	5								
B3	6	4								
C1	1	2								
C2	4	9								

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = 0$$

$\sqrt{(2-2)^2 + (10-5)^2} = 5$
 $\sqrt{(2-2)^2 + (5-5)^2} = 0$

K-Means Clustering – Solved Example

Initial Centroids:

A1: (2, 10)

B1: (5, 8)

C1: (1, 2)

Data Points			Distance to						Cluster	New Cluster
			2	10	5	8	1	2		
A1	2	10	0.00		3.61		8.06			
A2	2	5	5.00		4.24		3.16			
A3	8	4	8.49		5.00		7.28			
B1	5	8	3.61		0.00		7.21			
B2	7	5	7.07		3.61		6.71			
B3	6	4	7.21		4.12		5.39			
C1	1	2	8.06		7.21		0.00			
C2	4	9	2.24		1.41		7.62			

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Initial Centroids:

A1: (2, 10)

B1: (5, 8)

C1: (1, 2)

Data Points			Distance to						Cluster	New Cluster
			2	10	5	8	1	2		
A1	2	10	0.00		3.61		8.06		1	
A2	2	5	5.00		4.24		3.16		3	
A3	8	4	8.49		5.00		7.28		2	
B1	5	8	3.61		0.00		7.21		2	
B2	7	5	7.07		3.61		6.71		2	
B3	6	4	7.21		4.12		5.39		2	
C1	1	2	8.06		7.21		0.00		3	
C2	4	9	2.24		1.41		7.62		2	

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example



Initial Centroids:

A1: (2, 10)

B1: (5, 8)

C1: (1, 2)

New Centroids:

A1: (2, 10) ✓

B1: (6, 6) ✓

C1: (1.5, 3.5) ✓

Data Points			Distance to						Cluster	New Cluster
			2	10	5	8	1	2		
A1	2	10	0.00		3.61		8.06		1	
A2	2	5	5.00		4.24		3.16		3	
A3	8	4	8.49		5.00		7.28		2	
B1	5	8	3.61		0.00		7.21		2	
B2	7	5	7.07		3.61		6.71		2	
B3	6	4	7.21		4.12		5.39		2	
C1	1	2	8.06		7.21		0.00		3	
C2	4	9	2.24		1.41		7.62		2	

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

2. Distance

K-Means Clustering – Solved Example

Current Centroids:

A1: (2, 10)

B1: (6, 6)

C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			2	10	6	6	1.5	1.5		
A1	2	10	0.00		5.66		6.52		1	1
A2	2	5	5.00		4.12		1.58		3	3
A3	8	4	8.49		2.83		6.52		2	2
B1	5	8	3.61		2.24		5.70		2	2
B2	7	5	7.07		1.41		5.70		2	2
B3	6	4	7.21		2.00		4.53		2	2
C1	1	2	8.06		6.40		1.58		3	3
C2	4	9	2.24		3.61		6.04		2	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Current Centroids:

A1: (2, 10)

B1: (6, 6)

C1: (1.5, 3.5)

New Centroids:

A1: (3, 9.5) ✓

B1: (6.5, 5.25) ✓

C1: (1.5, 3.5) ✓

Data Points			Distance to						Cluster	New Cluster
			2	10	6	6	1.5	1.5		
A1	2	10	0.00		5.66		6.52		1	1
A2	2	5	5.00		4.12		1.58		3	3
A3	8	4	8.49		2.83		6.52		2	2
B1	5	8	3.61		2.24		5.70		2	2
B2	7	5	7.07		1.41		5.70		2	2
B3	6	4	7.21		2.00		4.53		2	2
C1	1	2	8.06		6.40		1.58		3	3
C2	4	9	2.24		3.61		6.04		2	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Current Centroids:

A1: (3, 9.5)

B1: (6.5, 5.25)

C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			3	9.5	6.5	5.25	1.5	3.5		
A1	2	10	1.12		6.54		6.52		1	1
A2	2	5	4.61		4.51		1.58		3	3
A3	8	4	7.43		1.95		6.52		2	2
B1	5	8	2.50		3.13		5.70		2	1
B2	7	5	6.02		0.56		5.70		2	2
B3	6	4	6.26		1.35		4.53		2	2
C1	1	2	7.76		6.39		1.58		3	3
C2	4	9	1.12		4.51		6.04		1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Current Centroids:

A1: (3, 9.5)

B1: (6.5, 5.25)

C1: (1.5, 3.5)

New Centroids:

A1: (3.67, 9) ✓

B1: (7, 4.33) ✓

C1: (1.5, 3.5) ✓

Data Points			Distance to						Cluster	New Cluster
			3	9.5	6.5	5.25	1.5	3.5		
A1	2	10	1.12		6.54		6.52		1	1
A2	2	5	4.61		4.51		1.58		3	3
A3	8	4	7.43		1.95		6.52		2	2
B1	5	8	2.50		3.13		5.70		2	1
B2	7	5	6.02		0.56		5.70		2	2
B3	6	4	6.26		1.35		4.53		2	2
C1	1	2	7.76		6.39		1.58		3	3
C2	4	9	1.12		4.51		6.04		1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Current Centroids:

A1: (3.67, 9)

B1: (7, 4.33)

C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			3.67	9	7	4.33	1.5	3.5		
A1	2	10	1.94		7.56		6.52		1	1
A2	2	5	4.33		5.04		1.58		3	3
A3	8	4	6.62		1.05		6.52		2	2
B1	5	8	1.67		4.18		5.70		1	1
B2	7	5	5.21		0.67		5.70		2	2
B3	6	4	5.52		1.05		4.53		2	2
C1	1	2	7.49		6.44		1.58		3	3
C2	4	9	0.33		5.55		6.04		1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$